

DSC602 June 2025 Exam - QUICK REFERENCE & STUDY GUIDE

Information Visualization | 70 Total Marks | 3 Hours

EXAM STRUCTURE

Section	Questions	Marks	Time
Q1: Reading & Comprehension	1.1-1.4	25	45 min
Q2: Data Description & Preparation	2.1-2.3	20	30 min
Q3: Graphics & Characteristics	6 graph types	25	45 min
TOTAL	3 questions	70	3 hours

QUESTION 1: READING & COMPREHENSION (25 marks)

Q1.1: Information Visualization Definition (6 marks)

Part A: What is InfoVis? (3 marks)

- Transformation of abstract, non-spatial data into visual forms
- Supports human perception, cognition, decision-making
- Key: NOT naturally spatial data (unlike maps)

Key Components (For full 3 marks):

1. **Input:** Abstract data (financials, text, relationships)
2. **Process:** Visual encoding (map data to visual properties)
3. **Output:** Interactive representations enabling insight

Part B: Meaning in Modern Context (3 marks)

1. Handles terabytes of data → makes invisible visible
2. Enables data-driven decision-making
3. Handles multi-dimensional complexity
4. Democratizes analytics to non-experts

✔ Model Answer Structure:

- Definition (1 mark)
- Key components explained (1.5 marks)
- Modern importance (0.5 marks)

Q1.2: Why Businesses Need Data Visualization (6 marks)

6 Key Reasons (1 mark each):

1. **Cognitive Advantage:** Humans process visuals 60,000× faster than text
2. **Pattern Recognition:** Visual cortex detects trends, outliers, clusters
3. **Decision Support:** Faster insights = quicker, better decisions
4. **Competitive Advantage:** Companies with better insights win
5. **Communication:** Replaces 50-page reports with one chart
6. **Handling Big Data:** Aggregation + interaction for massive datasets

✅ Answer Strategy:

- State the reason (0.5 mark)
 - Explain how it helps (0.5 mark)
 - Example: Business impact (optional, helps with clarity)
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Q1.3: Governance and Information Visualization (6 marks)**6 Applications of Visualization in Governance (1 mark each):**

1. **Performance Monitoring:** Track healthcare/education/finance metrics
2. **Budget Allocation:** Optimize resource distribution
3. **Policy Impact:** Evaluate if policies are working
4. **Citizen Engagement:** Transparency → public trust
5. **Identifying Issues:** Discover problems requiring intervention
6. **Benchmarking:** Compare with peer regions/nations

✅ Answer Pattern:

- Name application (0.3 mark)
- Specific use case (0.4 mark)
- Expected outcome (0.3 mark)

Real Examples to Mention:

- Singapore: Smart city dashboards
 - Denmark: Government spending transparency
 - USA: CDC disease tracking
 - UN: SDG (Sustainable Development Goals) monitoring
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Q1.4: Three Different Graphics from Class (6 marks)**Format: 3 graphs × 2 marks each****For EACH graph (2 marks total):**

1. **Description** (1 mark):
 - What it looks like (visual representation)
 - When to use it
 - Example

2. Why Useful (1 mark):

- 2-3 key advantages
- Business context

Common Options (Choose any 3):

Graph	Description	Why Useful
Line	Points connected by lines over time	Trends, seasonality, comparison
Bar	Rectangular bars representing values	Category comparison, simplicity
Histogram	Adjacent bins showing distribution	Distribution shape, outliers
Scatter	Points plotted on x-y axes	Correlation, relationships
Pie	Slices of circle (avoid if possible)	Part-to-whole only
Heatmap	Color intensity showing values	Multi-dimensional patterns

 Model Answer for 1 Graph (2 marks):

"The Bar Chart shows categorical data as rectangular bars proportional to values. In business, we use bar charts for sales by region - instantly see which regions perform best. This is useful because (1) heights are directly comparable, (2) can handle many categories, (3) easy for any audience to understand. Example: Our 5 regional sales can be compared in seconds, unlike tables which require manual scanning."

QUESTION 2: DATA DESCRIPTION & PREPARATION (20 marks)**Q2.1: Dictionary, Tuple, Series (9 marks)****Format: 3 data structures × 3 marks each****DICTIONARY (3 marks)****Definition** (1 mark):

- Unordered collection of key-value pairs: `{'key': value}`
- Mutable, flexible types

Characteristics (1 mark):

- Access by key (not position)
- Unordered (but insertion order maintained in Python 3.7+)
- Flexible types for keys and values

Data Viz Application (1 mark):

- Store configuration parameters
- Map category names to values
- Organize metadata

Code Example:

```
config = {'plot_type': 'bar', 'colors': ['red', 'blue'], 'title': 'Sales'}  
country_data = {'France': 85, 'Germany': 95, 'Italy': 62}  
# Used for flexible visualization parameter passing
```

TUPLE (3 marks)**Definition** (1 mark):

- Ordered, immutable collection: (element1, element2)
- Fixed size, cannot modify after creation

Characteristics (1 mark):

- Indexed access: tuple[0], tuple[1]
- Immutable (ensures data integrity)
- Hashable (can be dictionary keys)
- Lightweight, memory efficient

Data Viz Application (1 mark):

- Store coordinates (x, y), (x, y, z)
- Store RGB colors: (255, 128, 0)
- Store figure dimensions: (width, height)
- Return multiple values from functions

Code Example:

```
point = (10, 20) # 2D coordinate  
colors = [(255, 0, 0), (0, 255, 0)] # RGB tuples  
figure_size = (12, 8) # (width, height)
```

SERIES (3 marks)**Definition** (1 mark):

- Pandas 1D labeled array with index labels and values
- Built on NumPy, optimized for data operations

Characteristics (1 mark):

- Labeled (semantic meaning)
- Homogeneous (same type)
- Mutable, operations-rich
- Direct plotting capability

Data Viz Application (1 mark):

- Core data structure for visualization
- Direct `.plot()` method integration
- Filtering before visualization
- Time series with date indices

Code Example:

```
import pandas as pd
sales = pd.Series([100, 150, 120], index=['Q1', 'Q2', 'Q3'])
sales.plot(kind='bar', title='Quarterly Sales') # Direct plotting
```

Q2.2: Obtain, Maintain, Use for Visualization (6 marks)**Format: 3 structures × 2 marks each****DICTIONARY - Full 2 marks:**

- **Obtain** (0.5): Direct creation, from JSON, from DataFrame
- **Maintain** (0.5): Add/update with `d['key'] = value`, delete with `del d['key']`
- **Use** (1): Pass config to plotting functions, map labels to values

TUPLE - Full 2 marks:

- **Obtain** (0.5): Direct `(x, y)`, from list `tuple([...])`, unpacking `x, y = tuple_of_2`
- **Maintain** (0.5): Immutable - replace entirely, not modify
- **Use** (1): Store coordinates, colors, dimensions - ensures data integrity

SERIES - Full 2 marks:

- **Obtain** (0.5): Direct creation, from dict, from DataFrame column, from CSV
 - **Maintain** (0.5): Modify values, `fillna`, `dropna`, `reindex`
 - **Use** (1): Direct `.plot()`, filtering `series[series > threshold]`, time series
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Q2.3: Data Frame Particularity (5 marks)**Part A: What is a DataFrame? (3 marks)**

- 2D labeled structure (rows × columns)
- Each column is a Series
- Heterogeneous types (different columns different types)
- Index (labels rows), Column names (labels columns)

Definition (1 mark):

"A DataFrame is a 2-dimensional data structure with labeled rows and columns, where each column is a Series and data types can vary by column."

Structure (1 mark):

	Name	Age	Salary	Department
0	Alice	30	50000	IT
1	Bob	25	45000	HR
2	Carol	35	60000	IT

Key Features (1 mark):

- Labeled: Semantic meaning (column names, row indices)
- Mutable: Can add/remove rows/columns
- Indexed: Both label-based and position-based access
- Operations: Built-in methods for filtering, grouping, aggregation

Part B: Why Particularly Efficient for Programming (2 marks)**Efficiency 1: Vectorized Operations** (1 mark):

- Instead of: Manual loop over each row
- Use: `df['Bonus'] = df['Salary'] * 0.1` (one-liner!)
- 100x faster execution, cleaner code, automatic optimization

Efficiency 2: Built-in Methods Eliminate Code (1 mark):

- Grouping: `df.groupby('Dept')['Salary'].mean()`
- Filtering: `df[df['Age'] > 30]`
- Missing data: `df.fillna(df.mean())`
- All accomplished in single expressions vs. manual loops

**Model Answer (5 marks):**

"A DataFrame is a 2-dimensional labeled structure with rows and columns, where each column is a Series and can have different data types. It's particularly efficient for programming because: (1) Vectorized operations - `df['Bonus'] = df['Salary'] * 0.1` is 100x faster than looping, and (2) Built-in methods eliminate code - `df.groupby()`, `df.fillna()`, filtering with `df[df['Age'] > 30]` accomplish complex operations in single expressions. This makes data preparation extremely efficient."

QUESTION 3: GRAPHICS & CHARACTERISTICS (25 marks)**Overview**

Task: Identify which graph type is suitable for which context and justify.

6 Graph Types: Line, Bar, Histogram, Flow Chart, Box Plot, Bubble Map

Marks per Graph: 4 marks = (1) Definition + (2) Best Contexts + (1) Justification

GRAPH 1: LINE GRAPH (4 marks)

Definition (1 mark):

Connects data points with lines, showing change over time or ordered sequence.

Best Contexts (2 marks):

1. **Temporal Trends:** Stock prices, website traffic, patient health
2. **Multiple Series:** Revenue vs. Expenses over time, product comparison

Justification (1 mark):

- Pattern recognition: Trends visible instantly
- Seasonality detection: Cycles apparent
- Multiple series comparison
- Intuitive for general audiences

Example: Stock price chart showing 3 companies' performance over 100 days

GRAPH 2: BAR CHART (4 marks)**Definition** (1 mark):

Rectangular bars representing values, heights proportional to magnitudes.

Best Contexts (2 marks):

1. **Category Comparison:** Sales by region, market share by competitor
2. **Part-to-Whole (Stacked):** Revenue breakdown by product category

Justification (1 mark):

- Precise magnitude comparison
- Simplicity for any audience
- Multiple categories without clutter
- Composition analysis (stacked variant)

Example: Sales by 5 geographic regions with stacked breakdown by product

GRAPH 3: HISTOGRAM (4 marks)**Definition** (1 mark):

Distribution of continuous data using adjacent bins with frequency heights.

Best Contexts (2 marks):

1. **Distribution Analysis:** Age, income, test scores
2. **Outlier Detection:** Quality control, identify anomalies

Justification (1 mark):

- Distribution shape visible (normal, skewed, bimodal)
- Central tendency shown
- Outliers easily identified

- Data quality assessment

Example: Customer age distribution showing mostly 30-60 years with outliers

GRAPH 4: FLOW CHART (4 marks)

Definition (1 mark):

Diagram showing processes, decisions (diamonds), and flow (arrows) of steps.

Best Contexts (2 marks):

1. **Process Documentation:** Software workflows, customer service procedures
2. **Decision Trees:** Medical diagnosis, troubleshooting guides

Justification (1 mark):

- Sequential steps instantly clear
- Non-technical people understand
- Branching logic visible
- Identifies bottlenecks for optimization

Example: Customer service process with 3 branches (billing/tech/product)

GRAPH 5: BOX PLOT (4 marks)

Definition (1 mark):

Five-number summary visualization: min, Q1, median, Q3, max, with outliers.

Best Contexts (2 marks):

1. **Distribution Comparison:** Salary by department, test scores by school
2. **Outlier Identification:** Detect unusual measurements

Justification (1 mark):

- Compact multi-group comparison
- Statistical summary visible
- Outliers automatically marked
- Skewness detection (asymmetric boxes)

Example: Salary distribution across 4 departments with outlier identification

GRAPH 6: BUBBLE MAP (4 marks)

Definition (1 mark):

Geographic visualization with bubbles (circles) where size = magnitude, color = category.

Best Contexts (2 marks):

1. **Geographic Distribution:** Sales by city, population by region

2. **Multi-Dimensional:** Sales (size) by location + product type (color)

Justification (1 mark):

- Geographic context enables spatial pattern discovery
- Multiple dimensions (position + size + color)
- Intuitive for business stakeholders
- Identifies clusters and anomalies geographically

Example: Store locations with bubble size = sales, color = store type (Premium/Standard/Budget)

BONUS (1 mark)

- Overall integration/understanding
 - Clear presentation
 - Proper use of terminology
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STUDY TIPS & COMMON MISTAKES

DO THIS

1. **Be Specific:** Don't say "visualization is useful" - say WHY (with specific advantage)
2. **Provide Examples:** Abstract definitions are weak - use business examples
3. **Show Understanding:** Connect graph type to context and problem it solves
4. **Use Terminology:** "Central tendency", "outliers", "multi-dimensional", "vectorized"
5. **Structure Answers:** Use bullet points, clear sections, bold key terms
6. **Time Management:**
 - Q1: 45 minutes (6.25 min/part)
 - Q2: 30 minutes (10 min/section)
 - Q3: 45 minutes (7.5 min/graph)
 - Review: 5 minutes

AVOID THIS

1. **Generic Definitions:** "A line graph shows lines" - too vague
 2. **No Justification:** Don't list contexts without explaining WHY
 3. **Confusing Structures:** Dictionary vs Tuple - know the key difference (ordered? mutable?)
 4. **Missing Code:** For Q2, simple Python code examples help
 5. **Wrong Graph for Context:** Don't use pie charts, don't use line for categories
 6. **No Data Examples:** Theoretical without examples gets lower marks
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MARK ALLOCATION CHECKLIST

Q1 (25 marks)

- ☐ Q1.1a: Definition with components (3 marks)
- ☐ Q1.1b: Modern context importance (3 marks)

- ☐ Q1.2: 6 business reasons explained (6 marks)
- ☐ Q1.3: 6 governance applications (6 marks)
- ☐ Q1.4: 3 graphs with description + why useful (6 marks)
- **Subtotal: 25 marks**

Q2 (20 marks)

- ☐ Q2.1a: Dictionary - def + chars + app (3 marks)
- ☐ Q2.1b: Tuple - def + chars + app (3 marks)
- ☐ Q2.1c: Series - def + chars + app (3 marks)
- ☐ Q2.2: Obtain/Maintain/Use for each (6 marks)
- ☐ Q2.3: DataFrame def (3) + efficiency (2) (5 marks)
- **Subtotal: 20 marks**

Q3 (25 marks)

- ☐ Graph 1 (Line): 4 marks
- ☐ Graph 2 (Bar): 4 marks
- ☐ Graph 3 (Histogram): 4 marks
- ☐ Graph 4 (Flow): 4 marks
- ☐ Graph 5 (Box): 4 marks
- ☐ Graph 6 (Bubble): 4 marks
- ☐ Bonus/Integration: 1 mark
- **Subtotal: 25 marks**

TOTAL: 70 marks

KEY FORMULAS & FACTS

Data Structure Characteristics

Feature	Dictionary	Tuple	Series
Ordered	No*	Yes	Yes
Mutable	Yes	No	Yes
Access	By key	By index	By label/index
Type	Heterogeneous	Heterogeneous	Homogeneous
Example	<code>{'a': 1}</code>	<code>(1, 2)</code>	<code>pd.Series([1,2])</code>

Visualization Effectiveness Rule of Thumb

Cognitive Load: Visual ÷ Text = 60,000:1

- 1 chart ≈ 60,000 words
- Better than tables for patterns

Five-Number Summary (Box Plot)

- Min, Q1 (25%), Median (50%), Q3 (75%), Max
 - $IQR = Q3 - Q1$
 - Outliers = values $> Q3 + 1.5 \times IQR$ OR $< Q1 - 1.5 \times IQR$
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FINAL EXAM STRATEGY

Pre-Exam (Preparation)

- ☐ Memorize 3 graphs with definition, contexts, justification
- ☐ Practice defining data structures with Python examples
- ☐ Understand vectorized operations vs loops
- ☐ Know 3 governance examples

During Exam

- ☐ Read all questions first (5 min)
- ☐ Q1: Answer from stronger area first
- ☐ Q2: Use code examples for full marks
- ☐ Q3: Draw simple diagrams if describing graphs
- ☐ Review answers for terminology accuracy

Common High-Mark Patterns

- Specific examples from lecture or real business
 - Clear structure with bullet points
 - Theoretical explanation + practical application
 - Connection between concepts
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RESOURCES MENTIONED IN EXAM

Files Provided:

1. [DSC602_JUNE_2025_EXAM_SOLUTIONS.md](#) - Complete theory answers
 2. [DSC602_JUNE_2025_TITANIC_Q3_VISUALIZATIONS.ipynb](#) - All 6 graphs in Jupyter
 3. [DSC602_JUNE_2025_Q3_VISUALIZATIONS.py](#) - Python script version
 4. [DSC602_JUNE_2025_QUICK_REFERENCE.md](#) - This file
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LAST-MINUTE TIPS (15 minutes before exam)

1. **Dictionary vs Tuple:** Dict = flexibility (keys), Tuple = immutability (fixed)
 2. **Line vs Bar:** Line = TRENDS (time), Bar = COMPARISON (categories)
 3. **Histogram vs Box:** Histogram = DISTRIBUTION SHAPE, Box = GROUP COMPARISON
 4. **Why Visualization Matters:** Humans see patterns faster in visuals than numbers
 5. **Governance Example:** Use Singapore/Denmark/CDC examples if stuck
 6. **Flow Chart:** Non-technical people must understand (simplicity)
 7. **Bubble Map:** Size + Color + Position = 3 dimensions
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GOOD LUCK! You've got this! 

Remember: The exam tests your ability to:

- Define and explain concepts (theory)
- Connect concepts to business applications (context)
- Choose appropriate visualization for scenarios (judgment)
- Support arguments with examples (communication)